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# PUBLICATION LIST

## PATENTS

[P1] Reznik, L., & **Chuprov, S.** "Excluding Compromised Local Units from Receiving Global Parameters in Federated Learning Systems", WIPO Publication No. WO2024155819A1, published July 25, 2024.
*Filed by Rochester Institute of Technology; priority date January 19, 2023.*

[P2] Reznik, L., & **Chuprov, S.** "Network Adjustment Based on Machine Learning End System Performance Monitoring Feedback", WIPO Publication No. WO2024059625A1, published March 21, 2024.
*Filed by Rochester Institute of Technology; priority date September 14, 2022.*

## BOOKS & BOOK CHAPTERS

[B1] **Chuprov, S.**, Zatsarenko, R., & Reznik, L. (2025) "mLINK: Machine Learning Integration with Network and Knowledge" in Metacognitive Artificial Intelligence, eds. Shakarian, P. & Wei, H. Cambridge University Press, September 2025, Chapter 16, pp. 264-275. doi: 10.1017/9781009522472. URL

## REPRINTS

[R1] **Chuprov, S.**, Belyaev, P., Gataullin, R., Reznik, L., Neverov, E., & Viksnin, I. (2024) "Robust Autonomous Vehicle Computer-Vision-Based Localization in Challenging Environmental Conditions" in Applied Sciences. 2024, pp. 34-48., doi: doi.org/10.3390/books978-3-7258-2184-6. *Reprint of the Special Issue "Challenges in the Guidance, Navigation and Control of Autonomous and Transport Vehicles" that was published in Applied Sciences in 2023.* URL

## REFEREED JOURNAL PUBLICATIONS

[J1] Korobeinikov, D., Zatsarenko, R., **Chuprov, S.**, Barea, A., & Reznik, L. (2026). "InteFL Framework: Optimizing Federated Learning with Metacognition for Application Design and Deployment" in IEEE Intelligent Systems. 2026. doi: 10.1109/MIS.2026.3658072. URL. **Q1**. *Published in Special Issue*

[J2] **Chuprov, S.**, Reznik, L., & Zatsarenko, R. (2025). "KISS: Knowledge Integration System Service for ML End Attacks Detection and Classification" in IEEE Transactions on Dependable and Secure Computing. 2025, pp. 1-12. doi: 10.1109/TDSC.2025.3553807. URL. **Q1**.

[J3] **Chuprov, S.**, Zatsarenko, R., Reznik, L., & Khokhlov, I. (2024). "Data Quality Based Intelligent Instrument Selection with Security Integration" in ACM Journal of Data and Information Quality. 2024, vol. 16, no. 3, pp. 1-24. doi: 10.1145/3695770. URL. **Q2**.

[J4] **Chuprov, S.**, Belyaev, P., Gataullin, R., Reznik, L., Neverov, E., & Viksnin, I. (2023). "Robust Autonomous Vehicle Computer-Vision-Based Localization in Challenging Environmental Conditions" in Applied Sciences. 2023, no. 13(9):5735. doi: 10.3390/app13095735. URL. **Q1**. *Published in Special Issue*

[J5] Berezovskaya, O., **Chuprov, S.**, Neverov, E., & Sadreev, E. (2023). "Review and Comparison of Lightweight Modifications of the AES Cipher for a Network of Low-Power Devices" in Automatic Control and Computer Sciences. 2022, no. 56, pp. 994-1006. doi: 10.3103/S0146411622080028. URL

[J6] Melnikov, T., **Chuprov, S.**, Lazarev, E., Gataullin, R., & Viksnin, I. (2022). "Improving Reputation and Trust-Based Approach with Reliability Indicators for Autonomous Vehicles Intergroup Communication" in Tomsk State University Journal of Control and Computer Science. 2022, no. 61, pp. 108-117. URL

[J7] Marinenkov, E., **Chuprov, S.**, Tursukov, N., Kim, I., & Viksnin, I. (2022). "Study on Destructive Informational Impact in Unmanned Aerial Vehicles Intergroup Communication" in Symmetry. 2022. Vol. 14, no. 8, pp. 1-18. URL. **Q2**. *Published in Special Issue*

[J8] Viksnin, I., Marinenkov, E., & **Chuprov, S.** (2022). "A Game Theory approach for communication security and safety assurance in cyber-physical systems with Reputation and Trust-based mechanisms" in Scientific and Technical Journal of Information Technologies, Mechanics and Optics, vol. 22, no. 1, pp. 47–59. doi: 10.17586/2226-1494-2022-22-1-47-59. URL

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[J10] **Chuprov, S.**, Viksnin, I., Kim, I., Tursukov, N., & Nedosekin, G. (2020). Empirical Study on Discrete Modeling of Urban Intersection Management System. International Journal of Embedded and Real-Time Communication Systems (IJERTCS), vol. 11(2), pp. 16-38, doi: 10.4018/IJERTCS.2020040102. URL

[J11] **Chuprov, S.**, Viksnin, I., Kim, I., Marinenkov, E., Usova, M., Lazarev, E., Melnikov, T., & Zakoldaev, D. (2019). Reputation and Trust Approach for Security and Safety Assurance in Intersection Management System. Energies, 12(23), 4527, doi: 10.3390/en12234527. URL. **Q1**

## REFEREED CONFERENCES' PROCEEDINGS AND PRESENTATIONS

[C1] Amin, A., Hasan, K., Hong, L., **Chuprov, S.**, Thompson, R., Okpalaeze, A. D., Abena, P., Islam, T., & Ahmed, I. (2026). "Lightweight Privacy-First Federated Learning for Medical AI" in International Conference on Security and Privacy in Communication Systems (SecureComm 2026). doi: 10.1007/978-3-032-32764-2_13. URL

[C2] **Chuprov, S.**, Lange, R. D., Reznik, L., Shakarian, P., Zatsarenko, R., & Korobeinikov, D. (2026). "Position: Artificial Intelligence Needs Meta Intelligence — the Case for Metacognitive AI" in Proceedings of the 43rd International Conference on Machine Learning (ICML 2026). URL

[C3] Zatsarenko, R., Korobeinikov, D., **Chuprov, S.**, & Reznik, L. (2026). "Computationally Efficient Anomaly Detection and Exclusion for Practical and Robust Federated Learning" in IEEE/IFIP International Conference on Dependable Systems and Networks (DSN 2026).

[C4] Peña, A. & **Chuprov, S.** (2026). "Improving PID-Based Aggregation in Federated Learning for Consumer Applications" in 2026 IEEE 23rd Consumer Communications & Networking Conference (CCNC). 2026. doi: 10.1109/CCNC65079.2026.11366291. URL

[C5] Zatsarenko, R., **Chuprov, S.**, Korobeinikov, D., Reznik, L., & Peña, A. (2026). "How to Improve Federated Learning in Consumer Applications? Detect and Exclude Bad Clients" in 2026 IEEE 23rd Consumer Communications & Networking Conference (CCNC). 2026. doi: 10.1109/CCNC65079.2026.11366313. URL

[C6] Peredo, A. & **Chuprov, S.** (2026). "Improving Generalization with Harmonic Aggregation in Personalized Federated Learning" in 2026 IEEE 23rd Consumer Communications & Networking Conference (CCNC). 2026. doi: 10.1109/CCNC65079.2026.11366390. URL

[C7] Peña, A., **Chuprov, S.**, Zatsarenko, R., & Reznik, L. (2026). "Improving PID Control with Bayesian Optimization for Adversarially Robust Federated Learning" in Proceedings of the 9th International Conference on Information and Computer Technologies (ICICT 2026), pp. 149-154. doi: 10.1145/3803291.3803377. URL

[C8] Zatsarenko, R., **Chuprov, S.**, Korobeinikov, D., Reznik, L., & Peña, A. (2025). "MADE-PI: Framework for Effective Anomaly Detection in Federated Learning Applications" in 2025 International Conference on Machine Learning and Applications (ICMLA). 2025, pp. 446-451, doi: 10.1109/ICMLA66185.2025.00067. URL

[C9] Fonseca, P., **Chuprov, S.**, Solis, J. P., & Marquez, G. (2025). "Improving Lean Construction by Integrating ML for Contractor Performance Prediction" in 2025 38th Conference of Open Innovations Association (FRUCT). 2025. doi: 10.23919/FRUCT67853.2025.11239221. URL

[C10] Chan, E. E. N. & **Chuprov, S.** (2025). "Bayesian Optimization of PID Coefficients for Robust Federated Learning" in TACCSTER 2025 Proceedings. 2025. doi: 10.26153/tsw/62178. URL

[C11] Hong, L., **Chuprov, S.**, & Wan, Y. (2025) "Mitigating Health Misinformation Through Human-AI Collaboration" in AMCIS 2025 Proceedings. URL

[C12] **Chuprov, S.**, Korobeinikov, D., Zatsarenko, R., & Reznik, L. (2025) "No trust, no learning? Improving Federated Learning Security and Robustness with Reputation and Trust" in Data Science Workshop "From Theory to Practice". URL

[C13] Korobeinikov, D., **Chuprov, S.**, & Reznik, L., (2024). "Towards More Robust Federated Learning with Medical Imaging Model Anomaly Detection" in 2024 IEEE Western New York Image and Signal Processing Workshop (WNYISPW), 2024, pp. 1-4. URL

[C14] Narayanan, A., **Chuprov, S.**, Reznik, L., Zatsarenko, R., & Korobeinikov, D. "Intelligent Soccer Event Detection and Highlights Generation with Broadcast Cues Integration" in 23rd International Conference on Machine Learning and Applications (ICMLA'24), Miami, FL, USA, 2024, pp. 554-559, doi: 10.1109/ICMLA61862.2024.00081. URL

[C15] Patel, H., **Chuprov, S.**, Korobeinikov, D., Zatsarenko, R., & Reznik, L. (2024). "Improving Federated Learning Security with Trust Evaluation to Detect Adversarial Attacks" in 19th Annual Symposium on Information Assurance (ASIA' 24), 2024, pp. 37-43. URL

[C16] Korobeinikov, D., **Chuprov, S.**, Zatsarenko, R., & Reznik, L., (2024). "Federated Learning Robustness on Real World Data in Intelligent Transportation Systems" in 19th Annual Symposium on Information Assurance (ASIA' 24), 2024, pp. 30-36. URL

[C17] **Chuprov, S.**, Zatsarenko, R., Korobeinikov, D., & Reznik, L. (2024). "Robust Training on the Edge: Federated vs. Transfer Learning for Computer Vision in Intelligent Transportation Systems" in 2024 IEEE World AI IoT Congress (AIIoT), Seattle, WA, USA, 2024, pp. 172-178, doi: 10.1109/AIIoT61789.2024.10578970. URL

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[C21] **Chuprov, S.**, Mahajan, S., Zatsarenko, R., Reznik, L. & Ruchkan, A. (2023). "Are Industrial ML Image Classifiers Robust to Withstand Adversarial Attacks on Videos?" in 2023 IEEE Western New York Image and Signal Processing Workshop (WNYISPW), 2023, pp. 1-4, doi: 10.1109/WNYISPW60588.2023.10349595. URL

[C22] Zatsarenko, R., **Chuprov, S.**, Marathe, C.A., Hyland, M., & Reznik, L. (2023). "Are Industrial ML Image Classifiers Robust to Data Affected by Network QoS Degradation?" in 2023 IEEE Western New York Image and Signal Processing Workshop (WNYISPW), 2023, pp. 1-4, doi: 10.1109/WNYISPW60588.2023.10349560. URL

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[C25] **Chuprov, S.**, Bhatt, K.M., & Reznik, L. (2023). "Federated Learning for Robust Computer Vision in Intelligent Transportation Systems" in 2023 IEEE Conference on Artificial Intelligence (CAI), Santa Clara, CA, USA, 2023, pp. 26-27, doi: 10.1109/CAI54212.2023.00019. URL

[C26] **Chuprov, S.**, Memon, M., & Reznik, L. (2023). "Federated Learning with Trust Evaluation for Industrial Applications" in 2023 IEEE Conference on Artificial Intelligence (CAI), Santa Clara, CA, USA, 2023, pp. 347-348, doi: 10.1109/CAI54212.2023.00153. URL

[C27] **Chuprov, S.**, Reznik, L., & Grigoryan, G. (2022). "Study on Network Importance for ML End Application Robustness" ICC 2023 - IEEE International Conference on Communications, Rome, Italy, 2023, pp. 6627-6632, doi: 10.1109/ICC45041.2023.10279698. URL

[C28] **Chuprov, S.**, Satam, A. N., & Reznik, L. (2022). "Are ML Image Classifiers Robust to Medical Image Quality Degradation?" in 2022 IEEE Western New York Image and Signal Processing Workshop (WNYISPW), 2022, pp. 1-4, doi: 10.1109/WNYISPW57858.2022.9983488. URL

[C29] **Chuprov, S.**, Khokhlov, I., Reznik, L., & Manghi, K. (2022). "Multi-Modal Sensor Selection with Genetic Algorithms" in 2022 IEEE Sensors, 2022, pp. 1-4, doi: 10.1109/SENSORS52175.2022.9967296. URL

[C30] Khokhlov, I., **Chuprov, S.**, & Reznik, L. (2022). "Integrating Security with Accuracy Evaluation in Sensors Fusion" in 2022 IEEE Sensors, 2022, pp. 1-4, doi: 10.1109/SENSORS52175.2022.9967235. URL

[C31] **Chuprov, S.**, Khokhlov, I., Reznik, L., & Shetty, S. (2022). "Influence of Transfer Learning on Machine Learning Systems Robustness to Data Quality Degradation" in 2022 International Joint Conference on Neural Networks (IJCNN 2022), 2022, pp. 1-8, doi: 10.1109/IJCNN55064.2022.9892247. URL

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[C33] **Chuprov, S.**, Gataullin, R., Neverov, E., Belyaev, P., Kim, I., & Viksnin, I. (2022). "Police Office Model Performance and Security Evaluation in a Simulated Group of Mobile Robots" in The 5th International Conference on Future Networks & Distributed Systems (ICFNDS 2021). Association for Computing Machinery, New York, NY, USA, pp. 606–615, doi: 10.1145/3508072.3508195. URL

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[C35] Domnitsky, E., Mikhailov, V., Zoloedov, E., Alyukov, D., **Chuprov, S.**, Marinenkov, E., & Viksnin, I. (2021). "Software Module for Unmanned Autonomous Vehicle's On-board Camera Faults Detection and Correction" in CEUR Workshop Proceedings, Vol. 2893, pp. 1-10. URL

[C36] Lyakhovenko, Y., Viksnin, I., & **Chuprov, S.** (2021). "Integrating Smart Contracts into Smart Factory Elements' Informational Interaction Model" in CEUR Workshop Proceedings, Vol. 2893, pp. 1-6. URL

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[C39] Melnikov, T., Lazarev, E., Berezovskaya, O., **Chuprov, S.**, & Viksnin, I. (2020). "Empirical Study on Premises Monitoring Algorithm Implementation in Mobile Robotic System" in The International Conference "Nonlinearity, Information and Robotics", pp. 1-6, doi: 10.1109/NIR50484.2020.9290188. URL

[C40] Usova, M., **Chuprov, S.**, & Viksnin, I. (2020). "Informational Space and Messages Interaction Models for Smart Factory Concept" in 2020 IEEE International Workshop on Metrology for Industry 4.0 & IoT, pp. 617-621, doi: 10.1109/MetroInd4.0IoT48571.2020.9138292. URL

[C41] **Chuprov, S.**, Marinenkov, E., Viksnin, I., Reznik, L., & Khokhlov, I (2020). "Image Processing in Autonomous Vehicle Model Positioning and Movement Control" in IEEE 6th World Forum on Internet of Things (WF-IoT) Proceedings, pp. 1-6, doi: 10.1109/WF-IoT48130.2020.9221258. URL

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[C43] **Chuprov, S.**, Viksnin, I., & Kim, I. (2019) "Urban Intersection Management with Connected Infrastructure Objects and Autonomous Vehicles" in 2019 IEEE International Conference on Connected Vehicles and Expo (ICCVE), pp. 1-4, doi: 10.1109/ICCVE45908.2019.8964917. URL

[C44] **Chuprov, S.**, **Viksnin, I.**, **Kim, I.** & **Nedosekin, G.** (2019) "Optimization of Autonomous Vehicles Movement in Urban Intersection Management System" in 2019 24th Conference of Open Innovations Association (FRUCT),

pp. 60-66, doi: 10.23919/FRUCT.2019.8711967. URL

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[C47] Marinenkov, E., **Chuprov, S.**, Viksnin, I., & Kim, I. (2019). "Empirical Study on Trust, Reputation, and Game Theory Approach to Secure Communication in a Group of Unmanned Vehicles" in CEUR Workshop Proceedings, vol. 2590, pp. 1-12. URL

[C48] Usova, M., **Chuprov, S.**, Viksnin, I., & Baranova, O. (2019). "Model of Secure Informational Messages for Ensuring Informational Interaction in Smart Factory" in CEUR Workshop Proceedings, vol. 2590, pp. 1-8. URL

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[C50] Kim, I., Matos-Carvalho, J. P., Viksnin, I., Campos, L. M., Fonseca, J. M., Mora, A., & **Chuprov, S.** (2019). "Use of Particle Swarm Optimization in Terrain Classification based on UAV Downwash" in 2019 IEEE Congress on Evolutionary Computation (CEC), pp. 604-610, doi: 10.1109/CEC.2019.8790031. URL

[C51] Viksnin, I., **Chuprov, S.**, Usova, M., & Zakoldaev, D. (2019). "Police Office Model for Multi-agent Robotic Systems" in IOP Conference Series: Materials Science and Engineering, vol. 497, no. 1, p. 012036, doi: 10.1088/1757-899X/497/1/012036. URL

## DISSERTATIONS

[D1] **Chuprov, S.** (2024) "Robust Machine Learning Under Vulnerable Cyberinfrastructure and Varying Data Quality", Ph.D. dissertation, B. Thomas Golisano College of Computing and Information Sciences, Rochester Institute of Technology, Rochester, NY, USA, 2024. URL

[D2] **Chuprov, S.** (2019) "Assurance of Information in a Mobile Robotic System with Decentralized Control", Master's dissertation, Department of Secure Information Technologies, ITMO University, Saint Petersburg, Russia, 2019.

## OTHER PUBLICATIONS AND PRESENTATIONS

[O1] Fonseca, P., & **Chuprov, S.** (2025). "SPAM Email Detection App Using Machine Learning Techniques" in *UTRGV 2025 STEM Research Conference*. Presented May 2, 2025. — *Best Doctoral Poster Presentation in CS*

[O2] Belov, A., Belyaev, P., Viksnin, I., Kim, I., Radabolsky, V., Turushev, T., & **Chuprov, S.** (2023). "Software and Hardware Bundle for Controlling a Group of Unmanned Vehicles Based on Robust Computer Vision Algorithms" in LETI Transactions on Electrical Engineering & Computer Science. 2023, vol. 16, no. 6. pp. 52–69. – *in Russian.* URL

[O3] Berezovskaya, O., **Chuprov, S.**, Neverov, E., & Sadreev., E. (2022). "Review and Comparison of AES Lightweight Modifications for a Low-Power Devices Network" in Problems of Information Security. Computer Systems. no. 2, pp. 35-50. doi: 10.48612/jisp/gp9v-96dh-32
v1. – *in Russian.* URL

[O4] Buzina, E., **Chuprov, S.**, Tursukov, N., Belyaev, P., & Viksnin, I. (2022). "Algorithm for Uniform Coverage of the Monitoring Territory by Unmanned Aerial Vehicles" in LETI Transactions on Electrical Engineering & Computer Science. 2022. Vol. 15, no. 5/6. P. 41–50. doi: 10.32603/2071-8985-2022-15-5/6-41-50. – *in Russian.* URL

[O5] **Chuprov, S.**, Gataullin, R., & Viksnin, I. (2021). "Vulnerabilities Assesment in Mobile Robotic Control System" in Proceedings of the St. Petersburg State Electrotechnical University LETI, vol. 10, pp. 70-75. – *in Russian.* URL

[O6] **Chuprov, S.** (2020). "Intersection Management Tasks Application in Mobile Robotic System with Autonomous Vehicle Models" in Saint-Petersburg's Government Grant Award Winners, pp. 68-74. URL

[O7] **Chuprov, S.** (2020). "Reputation, Trust, and Data Quality Concept for Security and Safety Assurance in a Group of Autonomous Vehicles" in IX Young Scientists Congress Proceedings, Electronic Edition. URL

[O8] **Chuprov, S.** (2019). "Security Assurance of Mobile Robotic Systems with Decentralized Control" in 2019 ITMO University Annotated Master's Research Thesis Proceedings, pp. 66-72. URL

[O9] **Chuprov, S.**, Viksnin, I., & Kim, I. (2019). "Organization of Safe Intersections Passage by Unmanned Vehicles in a Mobile Robotic System" in VIII Young Scientists Congress Proceedings, Electronic Edition. URL

[O10] **Chuprov, S.**, Viksnin, I., Kim, I., & Usova, M. (2019). "Secure Intersection Management Passage with a Group of Autonomous Vehicles in Mobile Robotic System with Decentralized Control" in ITMO University Young Scientists Almanac, vol. 2, pp. 43-48. URL

[O11] **Chuprov, S.**, Viksnin, I., & Usova, M. (2018). "Police Office Model with Partial Decentralized Control in Mobile Robotic System" in Regional Informatics and Information Secutity SPOISU Proceedings, vol. 6, pp. 245-249. URL

**Presentations, posters, forums, workshops, and seminars w/o publications:**

[NP1] Fonseca, P., & **Chuprov, S.** (2025). "SPAM Email Detection App Using Machine Learning Techniques" in *STARTER-AI Summer 2025 Workshop*. Presented May 21, 2025.

[NP2] **Chuprov, S.** (2024). "About my Research, Approach to Mentorship, and Some Advice from my Fresh Grad Student's Perspective" at First-Year PhD Students in Computer Science seminar at UTRGV, 16 September 2024, Edinburg, TX, USA.

[NP3] **Chuprov, S.** (2024). "Security and Robustness of Machine Learning under Vulnerable Cyberinfrastructure and Varying Data Quality" at From Theory to Practice (T2P) Workshop, 15-19 April 2024, Kigali, Rwanda – *Remote presentation.*

[NP4] **Chuprov, S.**, Zatsarenko, R., & Reznik, L. (2024). "Improving Federated Learning Robustness towards Security Violations and Data Quality Degradations OR How to Learn Better via Extracting Knowledge and Accumulating History?" at UPSTAT 2024, 12-13 April 2024, Rochester, NY, USA.

[NP5] **Chuprov, S.**, & Reznik, L. (2023). "FLAME: Federated Learning against Malicious Engineering. Employing Trust and Reputation to Enhance Learning Security and Privacy" at Rochester Security Summit (RSS:2023), 25-26 October 2023, Rochester, NY, USA.

[NP6] **Chuprov, S.**, Reznik, L., & Zatsarenko, R. (2023). "MLIN or How to Make Networks and ML Applications Work Together in Real Conditions?" at First IEEE Upstate New York Workshop on Secure and Sustainable Communications Networks (SSCN), 10 October 2023, Rochester Institute of Technology, Rochester, NY, USA.

[NP7] **Chuprov, S.**, Reznik, L., & Zatsarenko, R. (2023). "FLAME: Federated Learning against Malicious Engineering. Employing Trust and Reputation to Enhance Learning Security and Privacy" at Eastern Great Lakes (EaGL) Theory Computation Workshop 2023, 30 September – 1 October 2023, University of Rochester, Rochester, NY, USA.

[NP8] **Chuprov, S.**, Reznik, L., & Memon, M. (2023). "Enhancing Federated Learning Security with Reputation and Trust-Based Indicators" at UPSTAT 2023, 21–22 April 2023, Rochester Institute of Technology, Rochester, NY, USA.

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